

SKILL - 3
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JupyterLab [?]
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[4]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
housing = fetch_california_housing(as_frame=True)
df = housing.frame
df.head()

#separate X and Y
X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]
#Train-Test Split "80% of the data is used for training and 20% for testing."
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("Training data:", X_train.shape)
#Linear Regression
lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)
#eval
rmse_lr = np.sqrt(mean_squared_error(y_test, y_pred_lr))
mae_lr = mean_absolute_error(y_test, y_pred_lr)
r2_lr = r2_score(y_test, y_pred_lr)
print("Linear Regression")
print("RMSE:", rmse_lr)
print("MAE :", mae_lr)
print("R2 :", r2_lr)
#Coefficients
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": lr.coef_
})
print(coefficients)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Training data: (16512, 8)
Linear Regression
RMSE: 0.7455813830127761
MAE : 0.5332001304956558
R2 : 0.5757877060324511

	Feature	Coefficient
0	MedInc	0.448675
1	HouseAge	0.009724
2	AveRooms	-0.123323
3	AveBedrms	0.783145
4	Population	-0.000002
5	AveOccup	-0.003526
6	Latitude	-0.419792
7	Longitude	-0.433708